George Sharpe

georgesharpe.dev

georgeasharpe@gmail.com
+44 7443 633232

Education

University of Exeter — Exeter, UK
BSc in Computer Science, 2:1 — *2022 – 2025*

Sandringham School — St Albans, UK
A-Levels — *2020 – 2022*

- **Grades**: Computer Science (A*), Maths (A*), Further Maths (B) & Physics (B)

Work Experience

Junior Data Engineer — Manchester, UK
Canopius Group — *Sep 2025 – Present*

- **Assumed ownership** of a legacy **SQL Server** & **Azure Data Factory** production project, rapidly gained domain expertise and become the **primary point of contact for technical guidance**.
  - Resolved all critical production failures **same-day** and prioritised non-urgent tasks alongside business stakeholders to maintain platform reliability.
  - Designed a novel algorithm in T-SQL independently to interpolate binder policy earned premiums, **reducing inaccuracy from 23% to 0%**. Validated results through Excel-based reconciliation. Used **SSMS, MDS**.
- Utilised **Terraform** to manage **Databricks** environments as **infrastructure as code**, **reducing outsourcing costs**. Deployed production updates to clear technical debt and deliver new features.
- Implemented **Databricks OpenSharing** via Terraform to securely ingest data from an external Australian partner, establishing reproducible infrastructure as code and **collaborating with cross-organisational stakeholders** to streamline data exchange.
- Managed release pipelines using **Azure DevOps** to track work items, execute deployments, and **approve PRs**, ensuring consistent software delivery, reduced deployment risk, and strict quality control across environments.
- Diagnosed and resolved critical production failures across ADF pipelines and SQL Server databases, conducting root-cause analysis via execution logs and queries to restore functionality with minimal downtime.
- **Proactively mentored and trained** new team members, including **knowledge-sharing sessions**, to establish internal expertise and enable seamless ongoing maintenance.
- Ingested and transformed JSON files using Databricks notebooks and **Delta Live Tables**, storing processed and unprocessed data in **Azure Data Lake** to enable reliable downstream analytics and reporting.
- Delivered BAU responsibilities by collaborating across colleagues and stakeholders, utilising **Kimball Data Warehousing** and **SQL Agent Jobs**, ensuring timely resolution of requests.

University Projects

- **Automated Clinical Coding - Python, Pandas, Transformers, Pytorch, Scikit-learn**:
  - Integrated clinical knowledge into **LLaMA-3** prompts using a **binary decision tree** for ICD code predictions.
  - Built an end-to-end clinical **NLP pipeline** by integrating data from three MIMIC-IV subsets for hierarchical diagnosis classification.
- **Event Management Database - SQL**:
  - Designed **ER diagrams** and translated to normalised **relational schemas** with justified constraints.
  - Basic **SQL** queries using joins and subqueries.
- **Customer Segmentation for E-Commerce - Python, Pandas, Scikit-learn, Matplotlib**:
  - Applied **K-Means clustering** with optimal k via Elbow method; performed feature selection and scaling.
  - Visualised clusters and inertia curves to communicate insights effectively.
- **Killer Sudoku Solver - Python, NumPy, SymPy**:
  - Created a novel algorithmic solver for the game Killer Sudoku by representing the game state as a matrix to then apply Gauss-Jordan elimination.

Skills

Languages: T-SQL, Python, Java, C#
Technologies: Terraform, Databricks, ADF, SSMS, MDS, SQL Agents Jobs, Kimball Data Warehousing, SSIS, Git